

8 (short)

(a) A uniform solid cylinder of length L is subjected to axial loading. Determine the nominal and true strains in the cylinder in the following two scenarios:

(i) the cylinder is stretched to twice its original length; [3]

(ii) the cylinder is compressed to half its original length. [3]

(b) Thermoplastic CD cases with weight between 25-35 g and a minimum section thickness of 1-2 mm are required. These cases should have surface roughness of $1\ \mu\text{m}$, and dimensional tolerance of 0.2 mm. The target batch size is 50,000. Using the relevant process attribute charts in the databook, identify the most suitable shaping process for the CD cases. [4]